



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 9 • STANDARD 6.AT.11

# Write Equations to Represent Relationships Between Two Variables

Lesson 9-3 • Teacher Copy

**Name:** Type your name

**Date:** Today's date

**Class:** Class period



## TODAY'S OBJECTIVES

**Content Objective:** I can write an equation from a table or a graph to model the relationship between the dependent and independent variables.

**Language Objective:** I can define the variables in my equation and explain what each part represents using the words equation, variable, and rate.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



**Fill in each blank as we go. Use the Word Bank to help you.**



## WORD BANK — FILL EACH BLANK WITH THE BEST WORD

equation

variable

rate

represent

define the variables

Constant



Tap any word to see its meaning and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

A mathematical sentence with an equals sign is an .

2

A letter standing for a quantity that can change is a .

3

A fixed amount of one quantity for every one unit of the other is a

.

4

To show a relationship with a table, graph, or equation is to

it.

5

To state what each letter stands for is to

.

6

A number in an equation that never changes is a  .

## Write About the Math

**Would you recommend that Diamond purchase or stream songs from the online music service?**

*¿Le recomendarías a Diamond comprar canciones o escucharlas en streaming del servicio de música en línea?*

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **equation**  
*ecuación*

☐ **variable**  
*variable*

☐ **rate**  
*tasa*

☐ **represent**  
*representar*

☐ **define the variables**  
*definir las variables*

**Model:** *The fee is charged EVERY month, so the total must be the rate multiplied by the number of months, not added to it once.* — A strong answer tests the equation against a table row (2 months should give \$49.90) rather than just stating a rule. A weak answer picks multiplication without checking it against the actual numbers.

### Start

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- **Diamond should write an \_\_\_\_ for each option so she can \_\_\_\_ the costs. Each equation uses the \_\_\_\_ as the input and gives the \_\_\_\_ as the output.**
- **My answer is \_\_\_\_ because \_\_\_\_.**

*Mi respuesta es \_\_\_\_ porque \_\_\_\_.*

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### Build

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
- **My evidence is \_\_\_\_, so \_\_\_\_.**

*Mi evidencia es \_\_\_\_\_, entonces \_\_\_\_\_.*

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### **Explain**

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

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☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

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## Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

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## Answer Key & Teacher Guide

1. **Notes 1:** equation
2. **Notes 2:** variable
3. **Notes 3:** rate
4. **Notes 4:** represent
5. **Notes 5:** define the variables
6. **Notes 6:** Constant
7. **Watch for this mistake:** *Writing the equation with addition —  $c = 24.95 + m$  — instead of multiplication. The fee is charged EVERY month, so the total is the rate times the number of months; always test the equation against a row of the table.*
8. **Try It 1:** D. The equation gives the total cost for ANY number of months, not just the rows listed — *A table can only show the values someone wrote down; substituting any value of  $m$  into  $c = 24.95m$  gives the cost for that many months instantly.*
9. **Try It 2:** A. 8 hours — *Divide both sides by 15:  $h = 120 \div 15 = 8$ .*